Features and Distractors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 8: know the target before judging a feature.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Choose recorded properties that preserve distinctions needed for a stated task. Try the recall before opening notes.

Your learning target

I can choose recorded properties that preserve distinctions needed for a stated task.

I can show why identical **feature record**s can hide examples with different **target label**s.

I can reject an answer copied into the **input**s and test a proposed **feature** with a new case.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

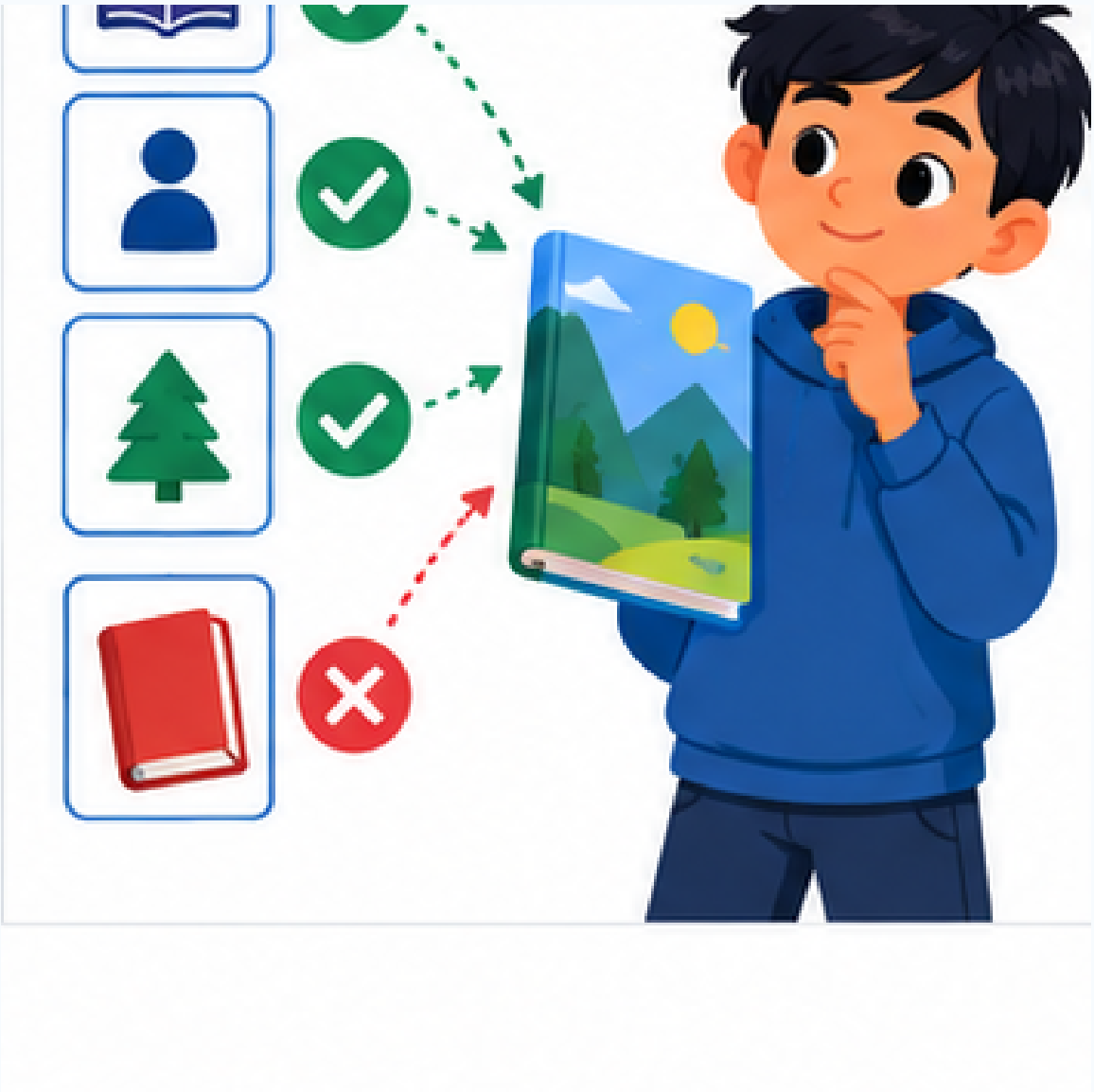
Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

After trying, check the **relevant** lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 9: **Feature**s and **Distractor**s

A **feature** should help the task and be available at decision time.



THINKER MISSION

Rank book **feature**s; reject cover color as a **shortcut**.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

This week's end goal

By the end, I can test a distracting **feature** with controlled changes rather than confounding clues.

STUDENT LESSON · 2 OF 18

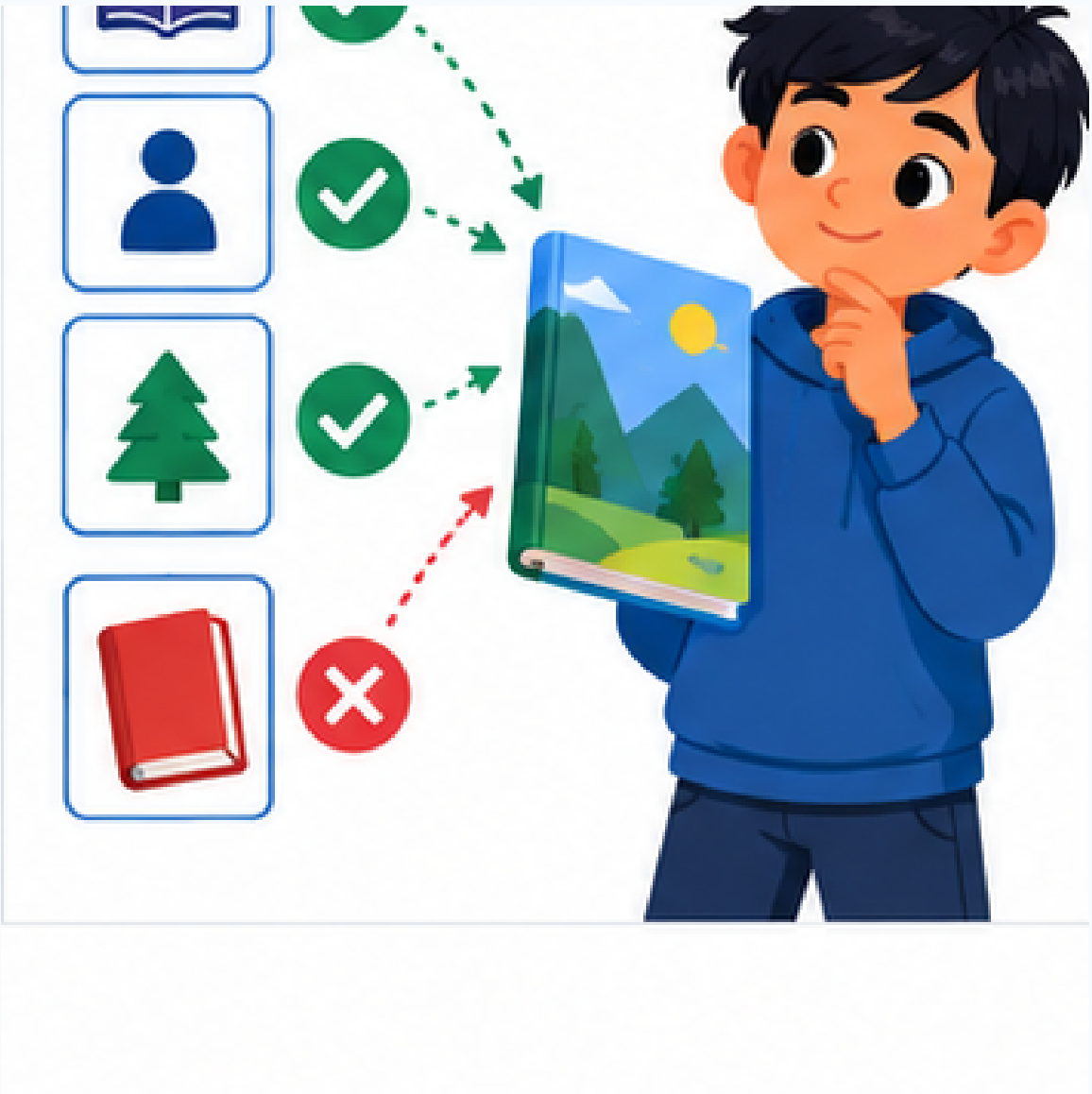
Notice the evidence

> Investigate the supplied Features and Distractors case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **feature**s and **distractor**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Features and Distractors. Connect each labelled input, deciding step and result.

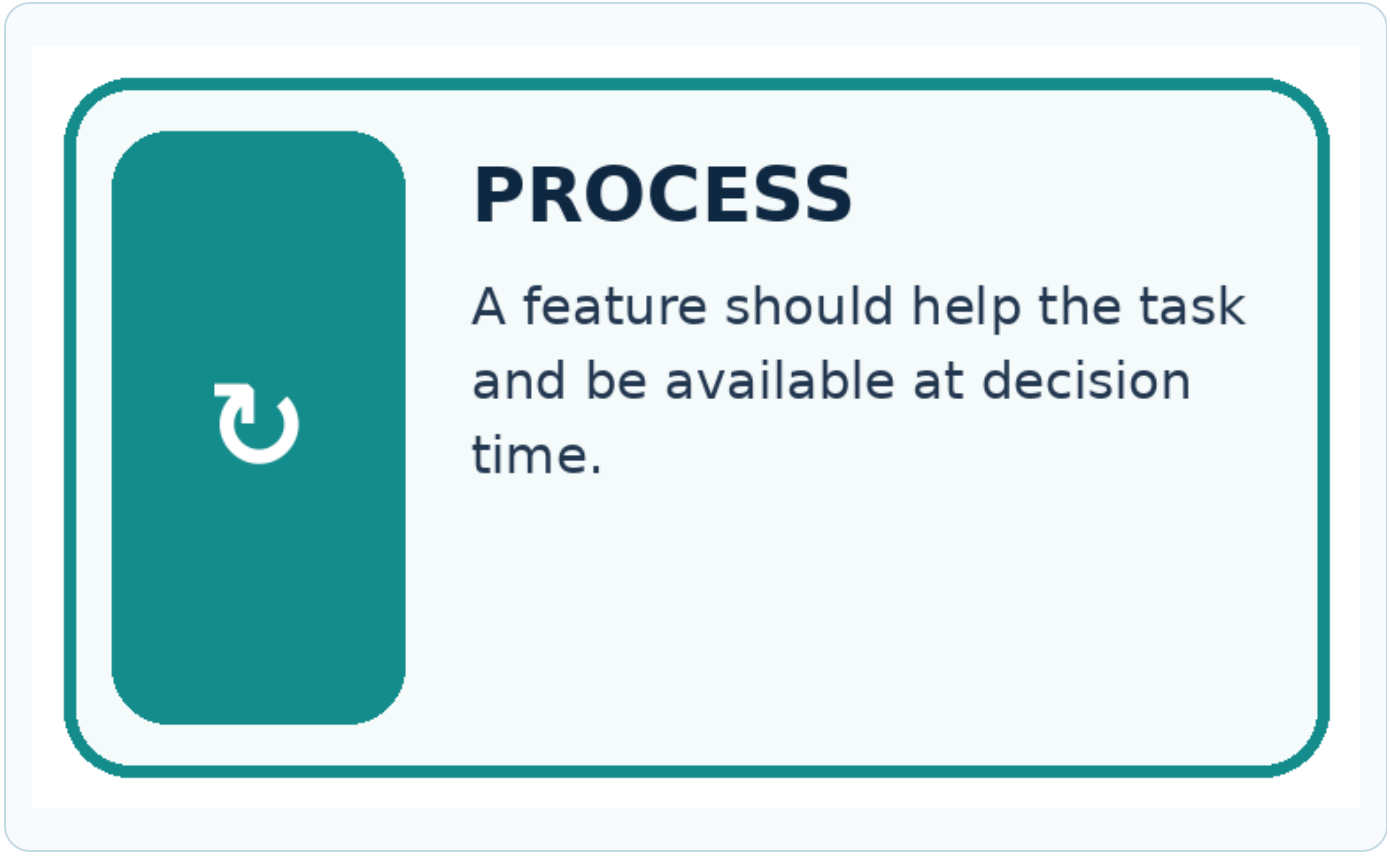
VISUAL MECHANISM

Input → Process → Output

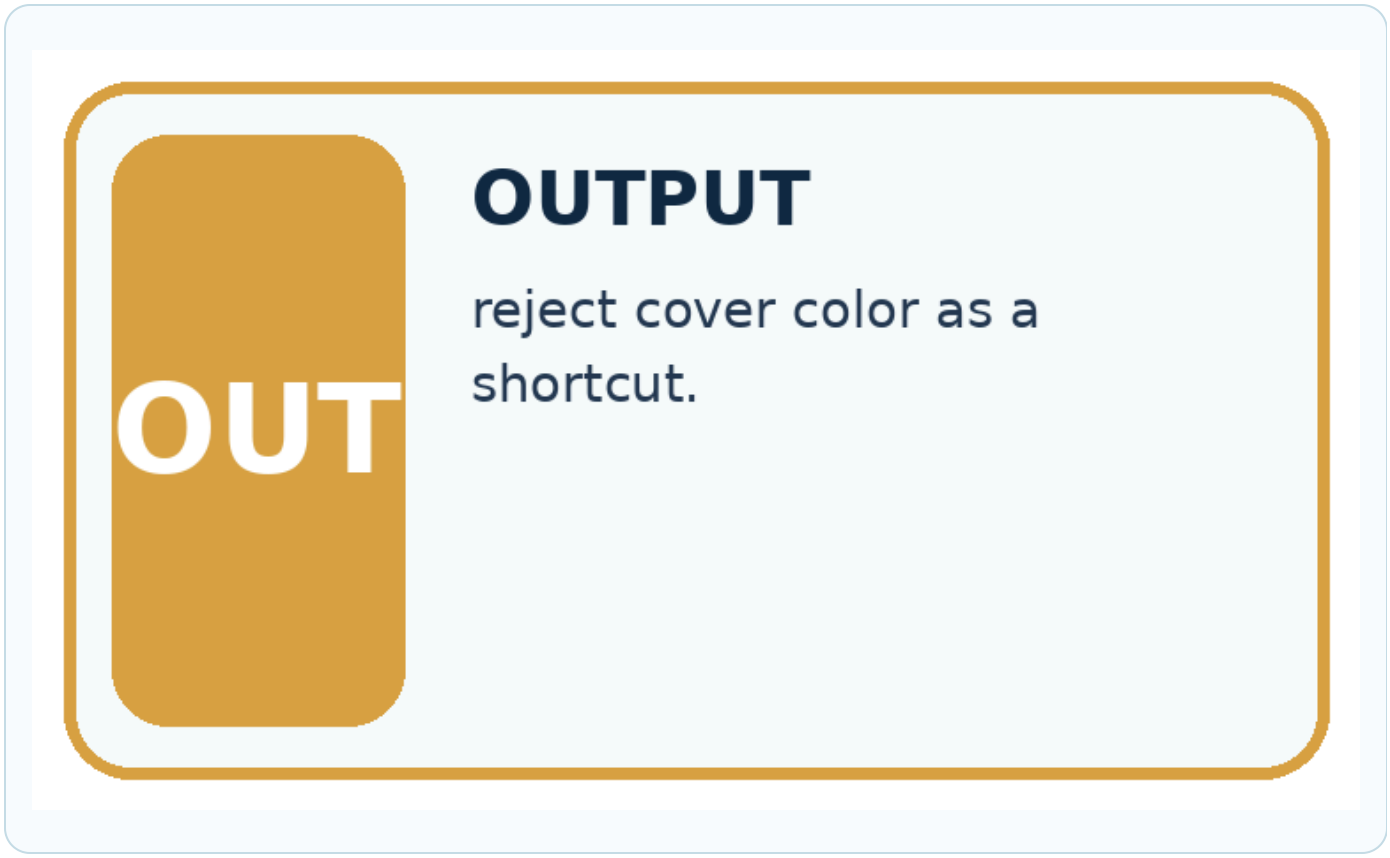
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A feature should help the task and be available at decision time.

↓


OUT
OUTPUT
reject cover color as a shortcut.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

The same cards under two feature choices

ORIGINAL CARD	CHOOSE THE INPUTS	RESULTING FEATURE RECORD
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

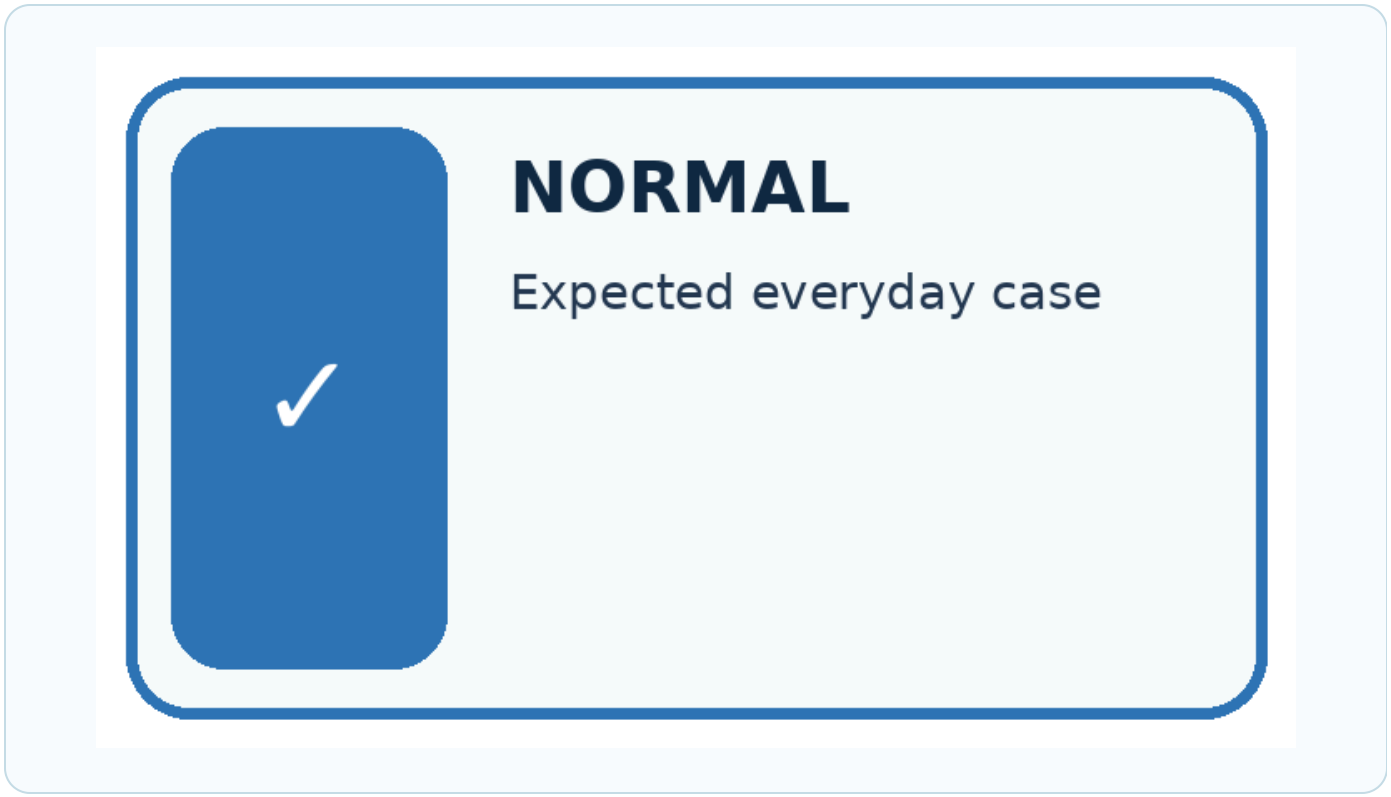
> Compare the supplied cases for Features and Distractors. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

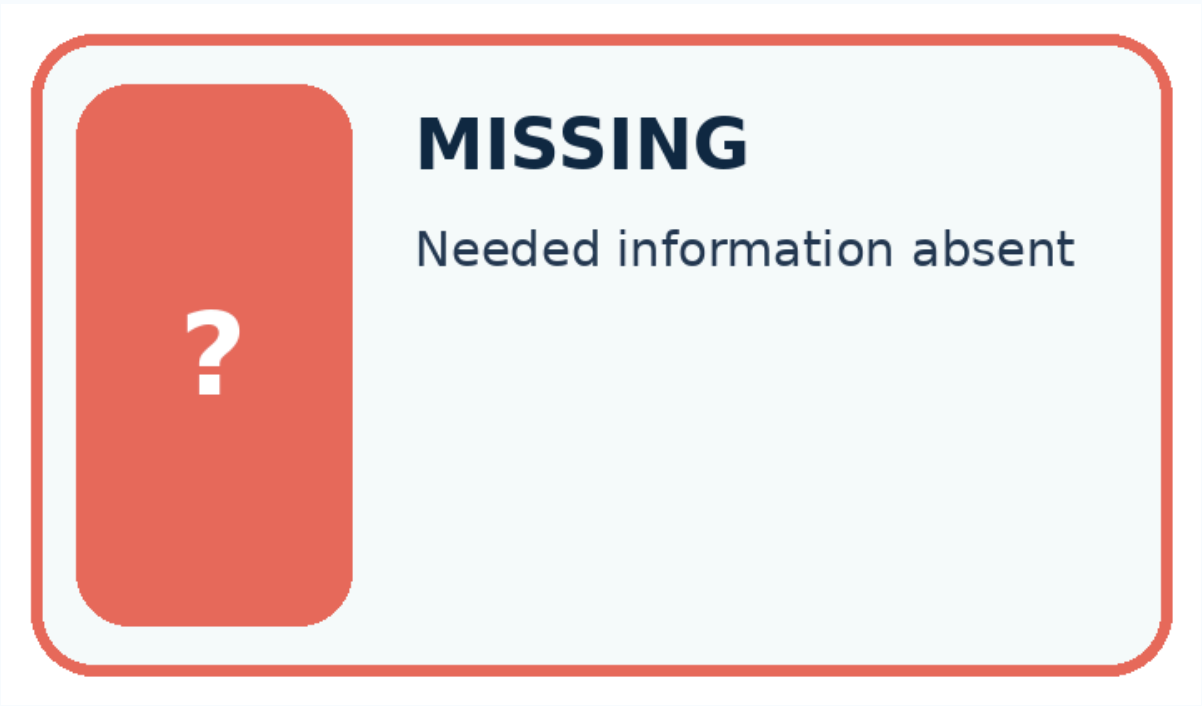
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The first controlled pair yields mystery for triangle and science for circle.

Predicted result

Prediction: That pair is **evidence** of unwanted logo sensitivity.

Why the result follows

Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

STUDENT LESSON · 5 OF 18

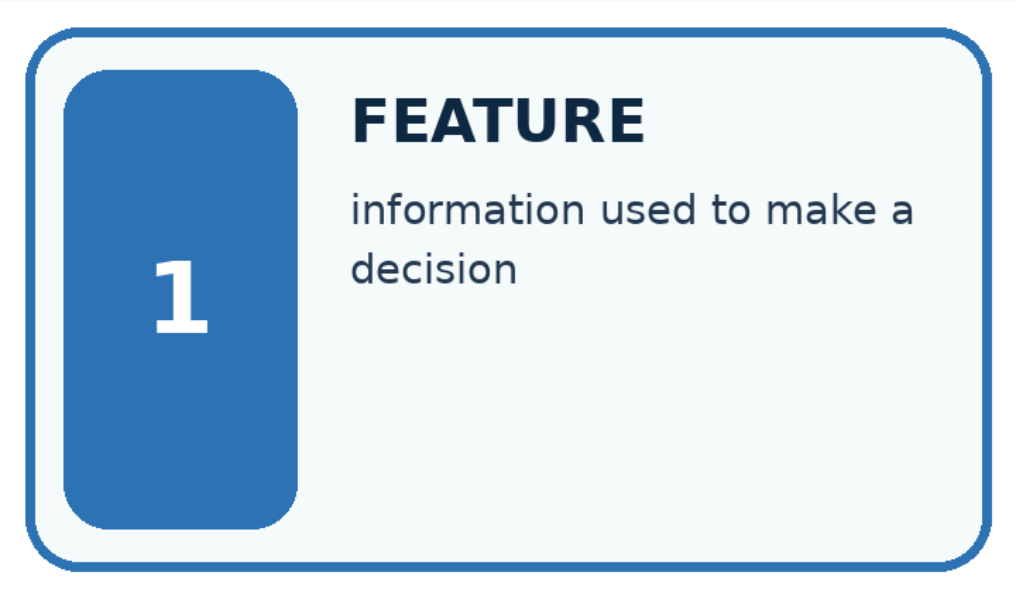
Words that unlock the idea

> Use the vocabulary of Features and Distractors to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

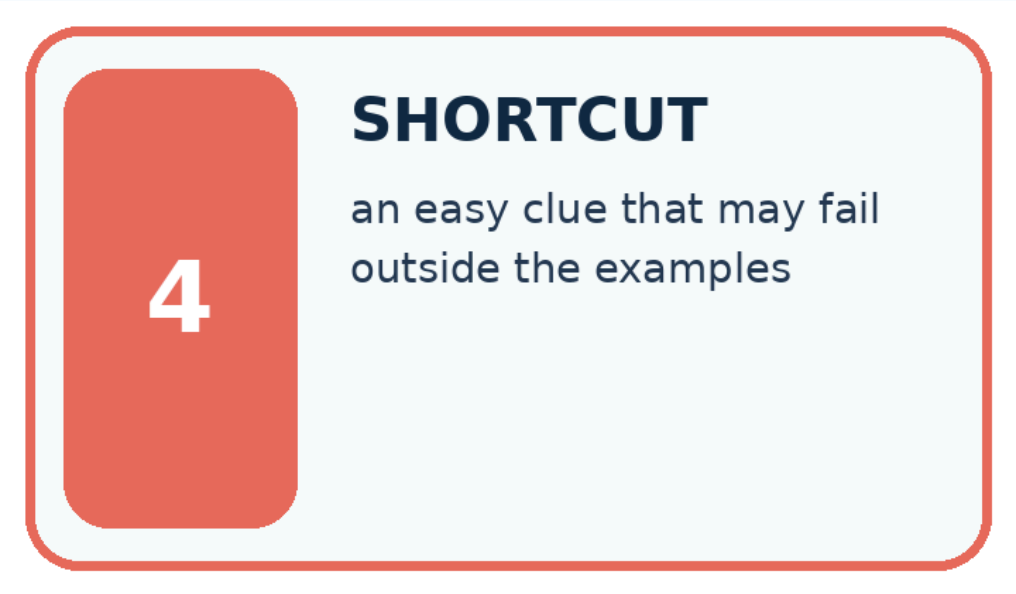
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
FEATURE
information used to make a decision


2
RELEVANT
connected to the goal





Word	My own example	Not this / boundary
feature	An input attribute or representation used by a model or decision rule.	
relevant	Useful to the particular question, target or decision being considered.	
distractor	A detail that draws attention but does not help answer the current question.	

Word	My own example	Not this / boundary
shortcut	A convenient but unreliable cue used instead of the relationship the task requires.	

Words used in the closer explanation

Word	Meaning
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Features and Distractors, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 9 • WORKED EVIDENCE

1. Goal: recommend enjoyable readable books.
2. Useful: genre match, reading level, length preference.
3. Distractor: cover color may match past labels by accident.
4. Deployment check: every chosen feature must exist before recommending.

1. Goal: recommend enjoyable readable books.

2. Useful: genre match, reading level, length preference.

3. **Distractor**: cover color may match past labels by accident.

4. Deployment check: every chosen **feature** must exist before recommending.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

Worked explanation · Features and Distractors

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Worked explanation · Features and Distractors

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the **shortcut**.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features and Distractors

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

The facts we will use

Supplied facts

Training contains no mystery-circle or science-triangle covers.

Supplied facts

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

AI ACADEMY / AI-2/9

Features and Distractors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Spot confounding

Topic and logo always move together in the training set.



2. Hold topic fixed

Compare the same mystery description with triangle versus circle logo.



3. Hold logo fixed

Compare mystery versus science descriptions with identical triangle logo.



4. Observe separately

Record predictions before concluding which clue the classifier uses.

CHANGE ONE THING

The first controlled pair yields mystery for triangle and science for circle.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Spot confounding	Topic and logo always move together in the training set.
2. Hold topic fixed	Compare the same mystery description with triangle versus circle logo.
3. Hold logo fixed	Compare mystery versus science descriptions with identical triangle logo.
4. Observe separately	Record predictions before concluding which clue the classifier uses.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

What this example does not establish

Keep the stated limit

Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

Keep the stated limit

Read the labelled picture: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

Keep the stated limit

Change one thing: The first controlled pair yields mystery for triangle and science for circle. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Features and Distractors claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Any field correlated with past choices is automatically safe to use.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Before adopting a **feature** set, state the task, name the **input**s with units, and ask whether two different answers can share the same **input** record. Construct such a pair if you can.

Then check that each **input** is available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

Check the claim against the facts

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly.

That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the **feature**s again.

More columns are not automatically better; each one needs a reason to be collected and used.

Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead

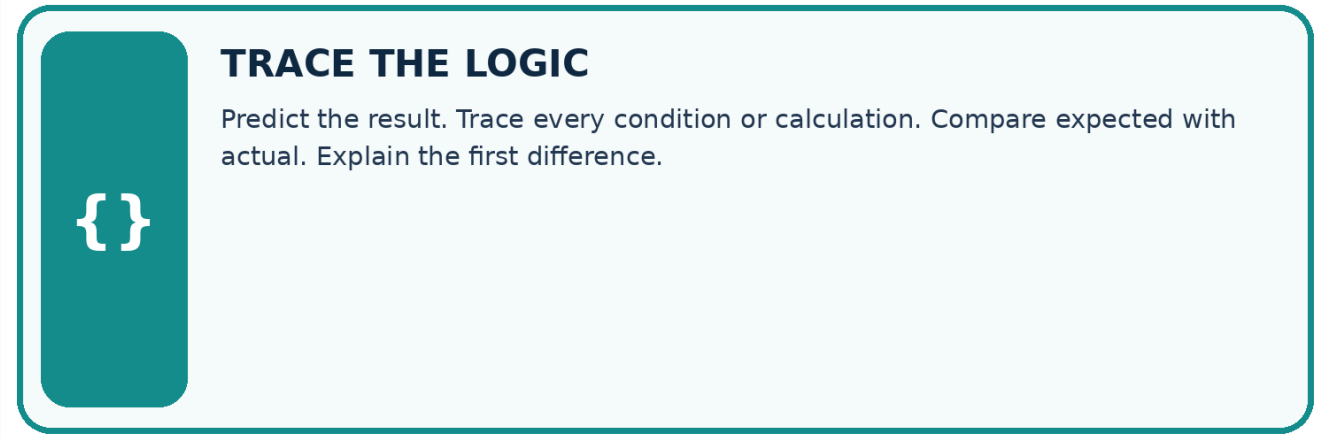
Practise together

> Practise Features and Distractors with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an **input** to predict the label before that inspection.

The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended **prediction**. This is **leakage**: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker.

Keep labels available for teaching and checking **prediction**s, but do not feed the sticker back in as an **input** for this task.

> **A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label**

Practise with less help: Prove that one measurement is insufficient

Use workbook W2 for the connected example "**Feature**s and useful distinctions". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: "A stronger model can always recover an omitted measurement". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

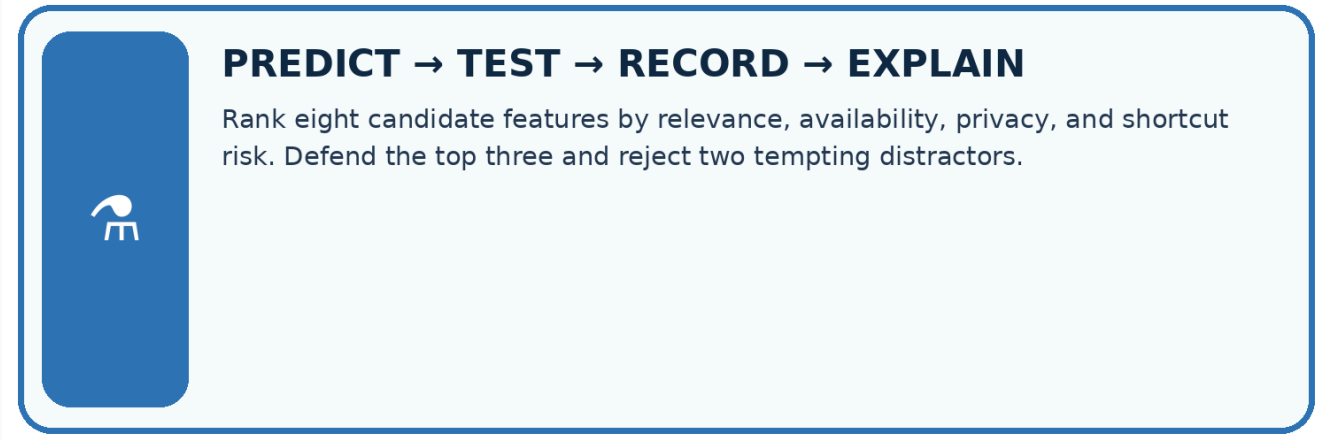
Run the investigation

> Investigate Features and Distractors. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Rank eight candidate **feature**s by relevance, availability, privacy, and **shortcut** risk. Defend the top three and reject two tempting **distractor**s.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

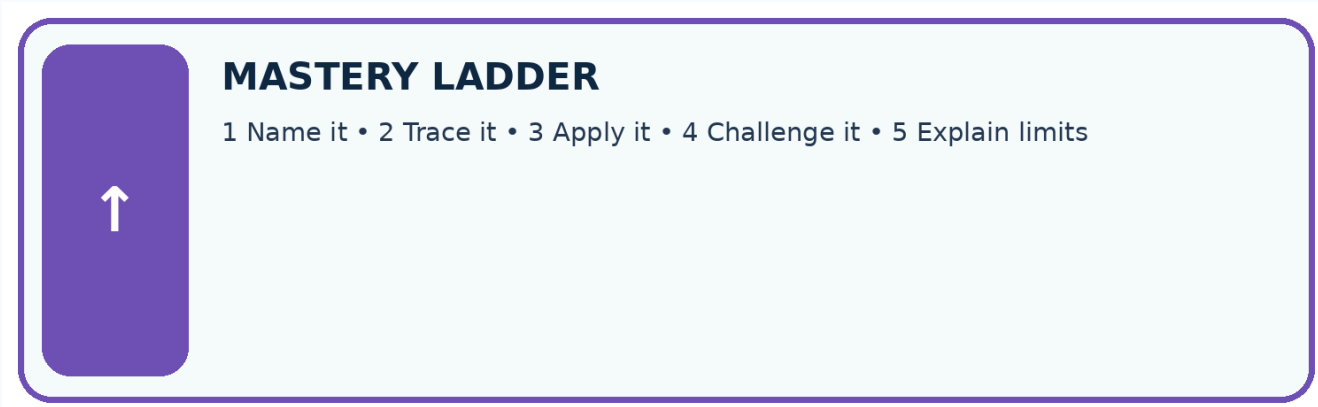
Build your understanding

Move from recall to tracing, diagnosis and transfer in Features and Distractors. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

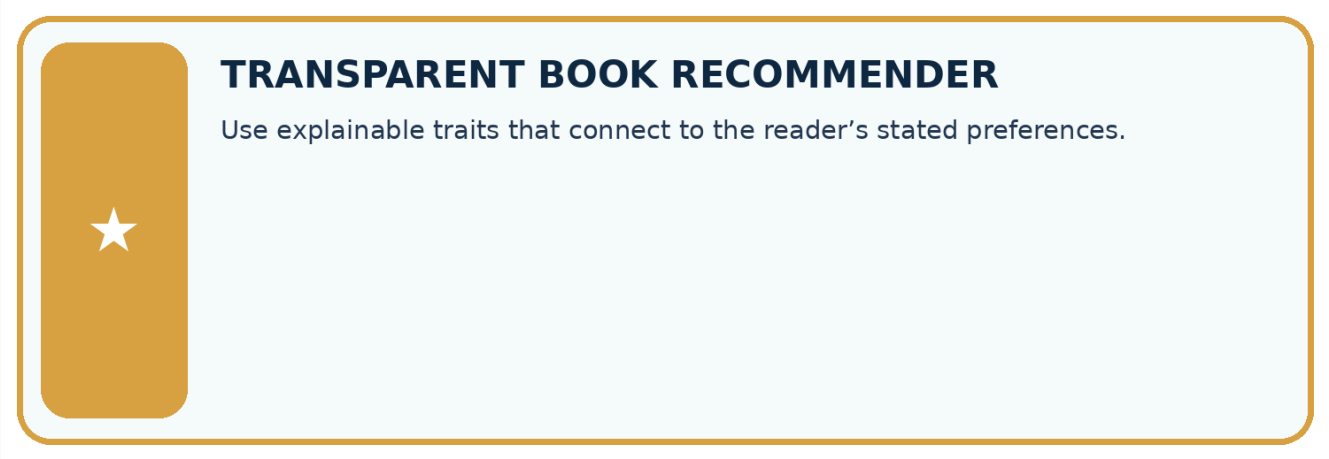
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Use explainable traits that connect to the reader's stated preferences.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Rank book **feature**s; reject cover color as a **shortcut**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Features and Distractors case for supported practice. Repeated examples do not become fresh assessments.

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

The question

Choose useful **evidence** and possible distractions.

Worked explanation

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Useful evidence versus a shortcut

Evidence or stage	Value or result
Target	Contains animal characters
Animal tag	Relevant if accurate

Evidence or stage	Value or result
Cover colour	Possible distraction
Shelf number	Possible accidental shortcut

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Training covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Predict whether a book is short using page count and cover colour. Which **feature** directly supports the defined label? Give a counterexample to colour.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one **relevant** detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use explainable traits that connect to the reader's stated preferences.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Features and Distractors, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

By the end, I can	How I will show it
Choose recorded properties that preserve distinctions needed for a stated task	Explain the deciding step, show a trace or test, and state one limit.
Show why identical feature records can hide examples with different target labels	Explain the deciding step, show a trace or test, and state one limit.
Reject an answer copied into the inputs and test a proposed feature with a new case	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Features and Distractors without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose recorded properties that preserve distinctions needed for a stated task

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record **prediction**s before concluding which clue the classifier uses.

1. Hold topic fixed

2. Hold logo fixed

3. Observe separately

4. Spot confounding

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Observe separately Record **prediction**s before concluding which clue the classifier uses.

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The first controlled pair yields mystery for triangle and science for circle. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. That pair is **evidence** of unwanted logo sensitivity.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That pair is **evidence** of unwanted logo sensitivity. The subject description stayed mystery; only the ir**relevant** logo changed.

Repair a missing building block

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

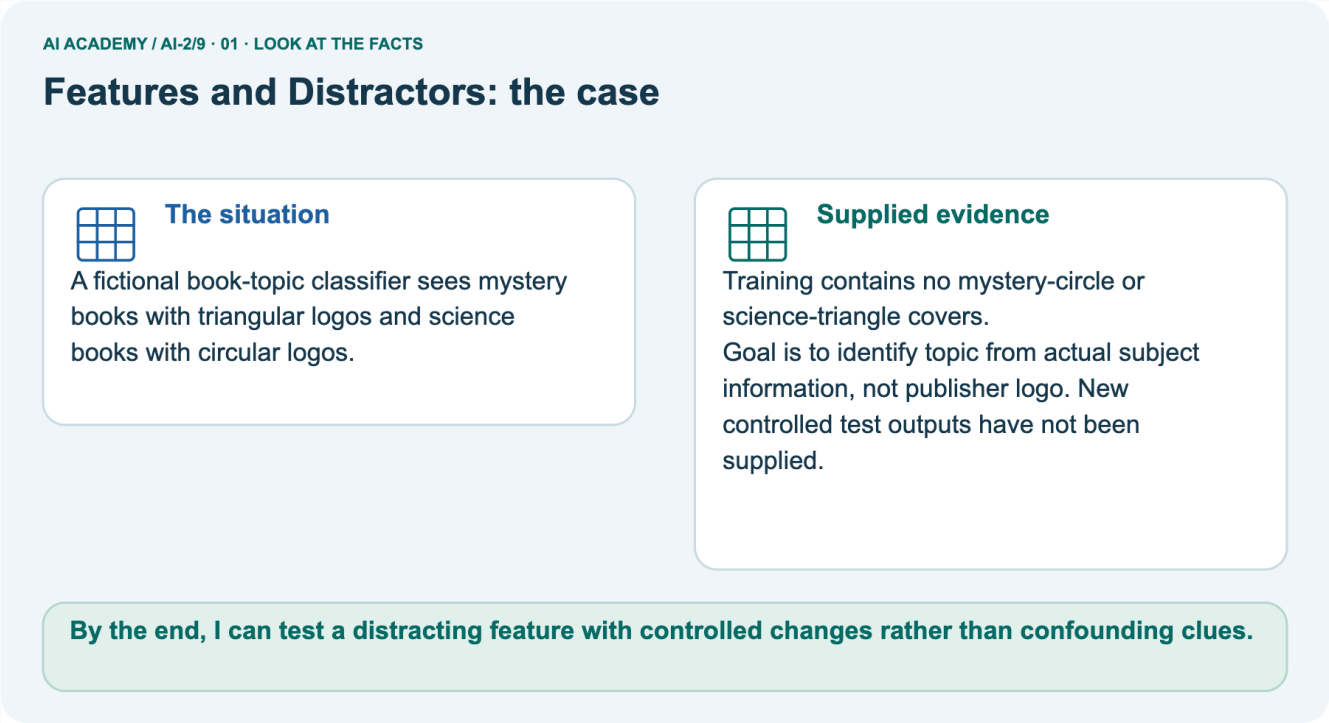
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Features and Distractors graphic using its labels, deciding evidence and relevant limitation.

01. Features and Distractors: the case



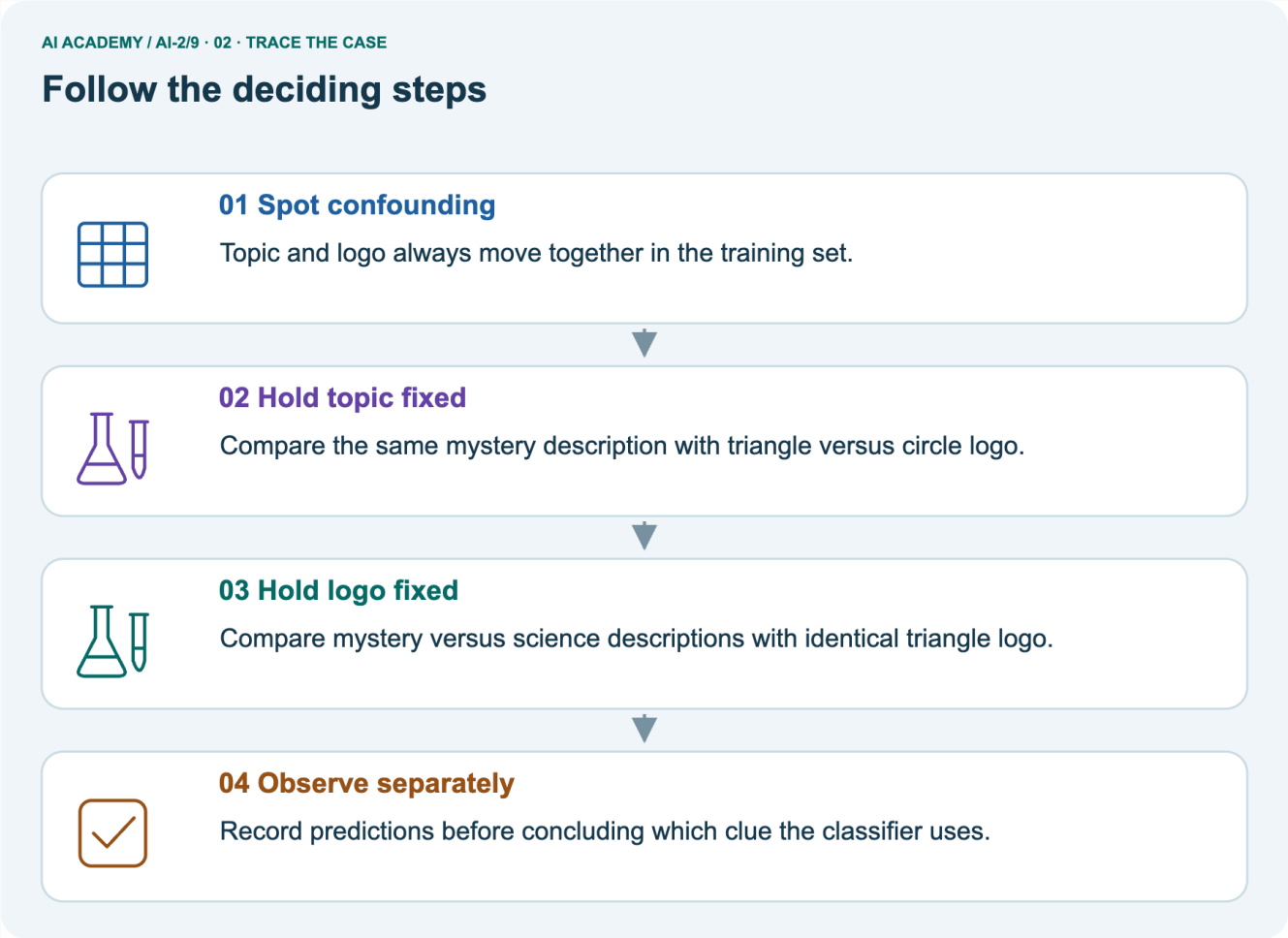
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos. Supplied facts: **Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

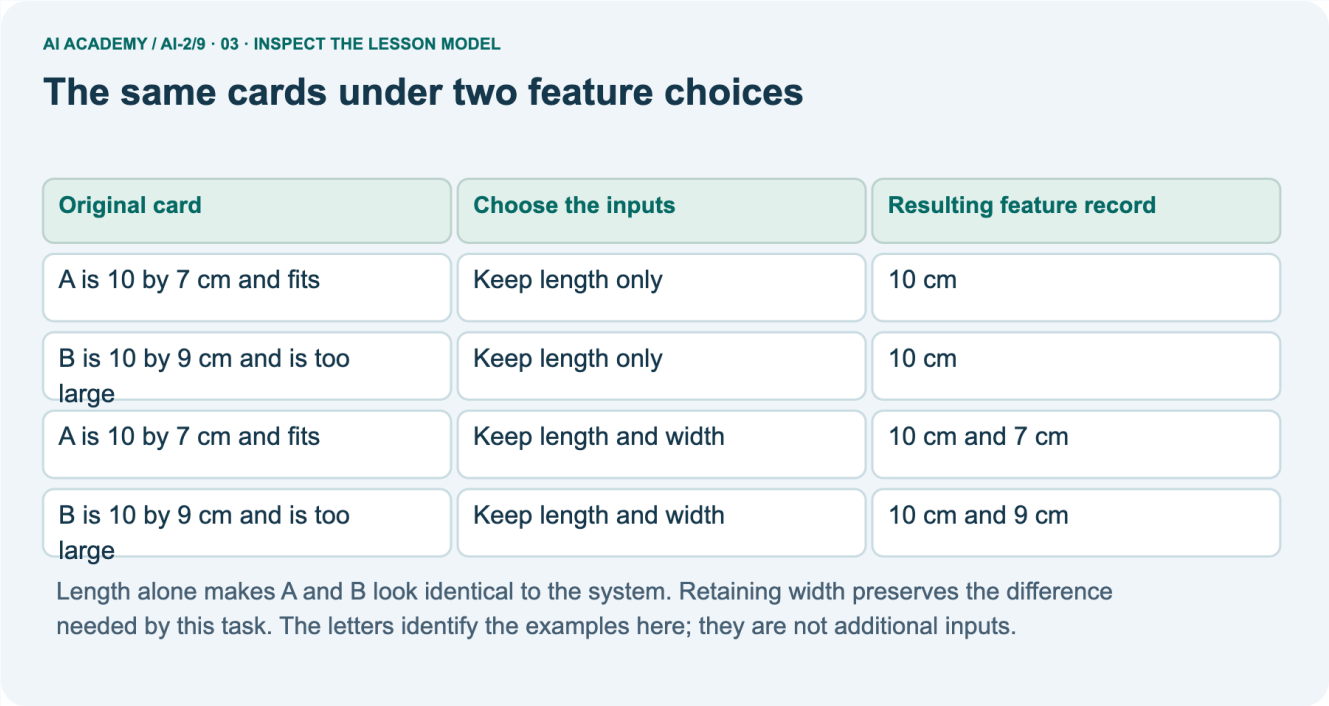


Follow the labelled picture with the matching explanation. Open the vector diagram

Spot confounding: Topic and logo always move together in the **training** set. → Hold topic fixed: Compare the same mystery description with triangle versus circle logo. → Hold logo fixed: Compare mystery versus science descriptions with identical triangle logo. → Observe separately: Record **prediction**s before concluding which clue the classifier uses.

> Point to each numbered stage and explain which recorded **input** it uses.

03. The same cards under two feature choices



AI ACADEMY / AI-2/9 · 03 · INSPECT THE LESSON MODEL

The same cards under two feature choices

Original card	Choose the inputs	Resulting feature record
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

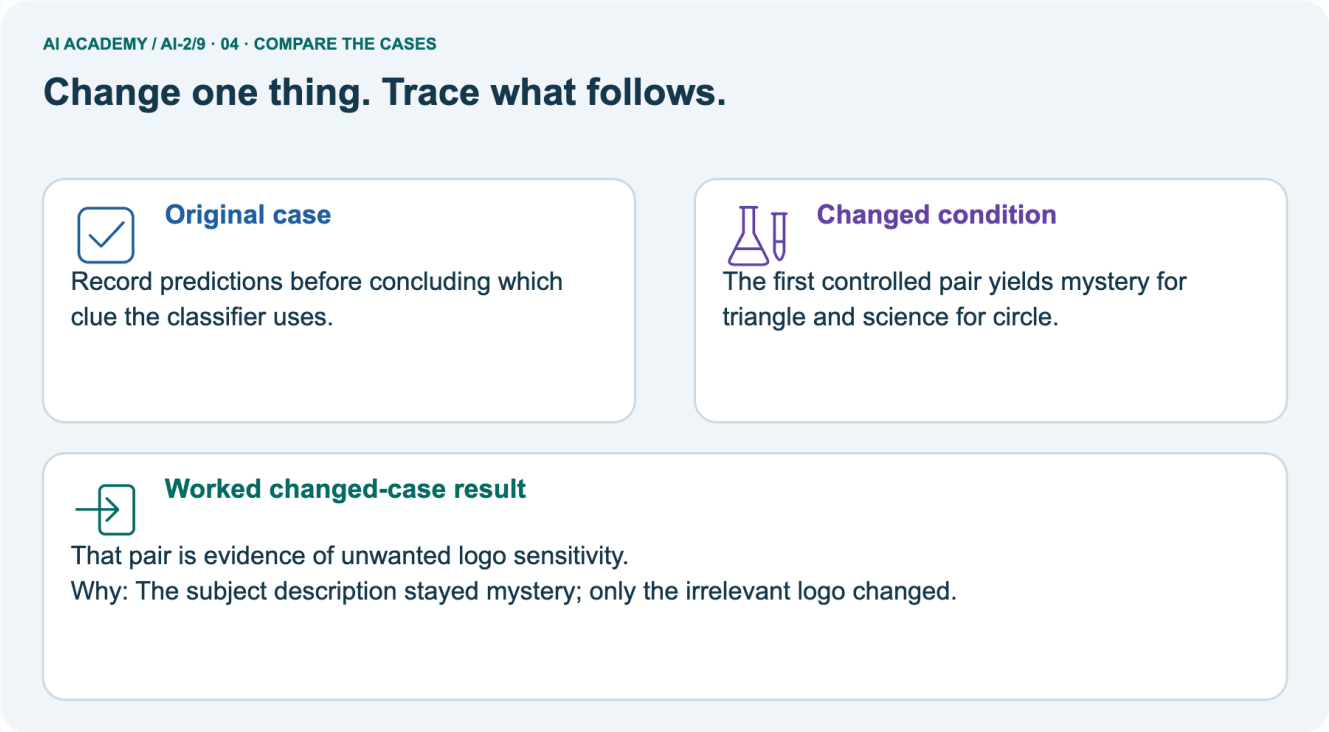
Follow the labelled picture with the matching explanation. Open the vector diagram

The same cards under two **feature** choices. A is 10 by 7 cm and fits | Keep length only | 10 cm. B is 10 by 9 cm and is too large | Keep length only | 10 cm. A is 10 by 7 cm and fits | Keep length and width | 10 cm and 7 cm.

B is 10 by 9 cm and is too large | Keep length and width | 10 cm and 9 cm

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



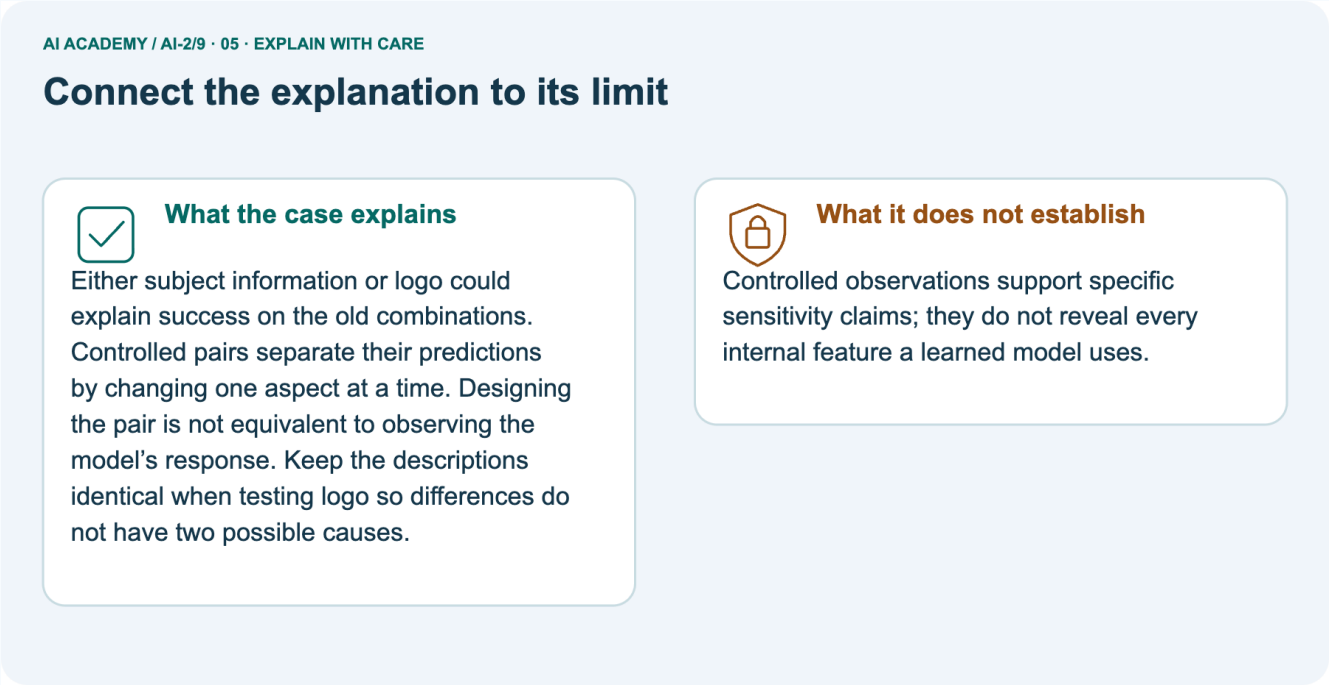
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **prediction**s before concluding which clue the classifier uses. Changed condition: The first controlled pair yields mystery for triangle and science for circle.

Expected result: That pair is **evidence** of unwanted logo sensitivity. Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

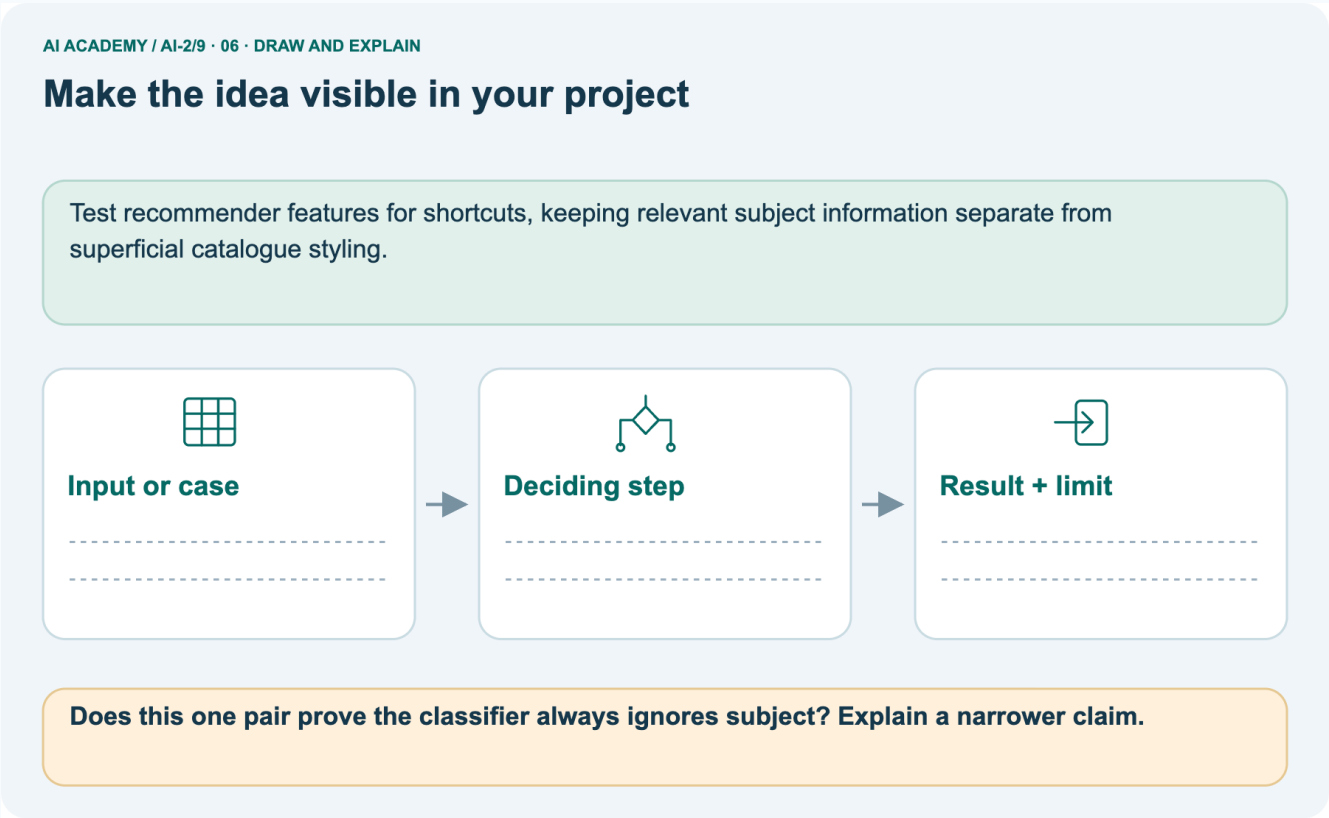
Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Limit: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling. Draw your **input**, deciding step, result and limit.

Supported question: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.